

Flood Risk Classifier — Defense Guide

A plain-language companion to `Flood_Risk_Classifier_Model_Report.pdf`. Read this first to get the whole model in your head quickly; go to the full report only for the literature tables and appendix.

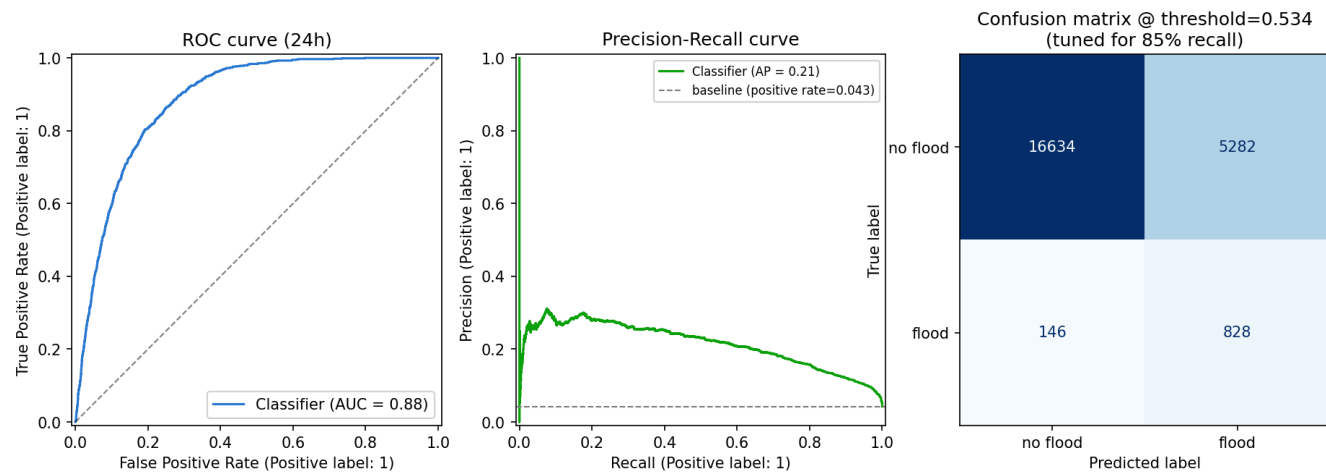
1. The 60-second summary

This model answers one question: **will this station flood in the next 24, 48, or 72 hours?**, using today's live rainfall, soil moisture, and river discharge. It's not one model — it's **three independent LightGBM classifiers**, one per horizon, each trained across all 30 monitored stations at once (not one model per station). Each one outputs a flood probability; a tuned threshold turns that into a low/moderate/high risk level.

The single most important thing to know walking in: the model **beats a "no rain expected" climatology baseline clearly, but does not beat a "tomorrow will look like today" persistence baseline** on the metric that matters most (PR-AUC). This is reported honestly in the full report (§8), not hidden — and it has a credible, stated explanation (see Q6 below).

2. How it works, in plain terms

1. **Inputs, live, every request:** rainfall at the station and its upstream catchment, soil moisture, and current river discharge — pulled from free public APIs (Open-Meteo), not paid data.
2. **A borrowed signal:** each horizon's classifier also gets that same horizon's own discharge forecast (from the companion discharge-forecasting model, bundled inside this model's own `models/discharge_cascade/`) as an extra input — tested to genuinely help, not just added on faith.
3. **Three separate models, not one:** a 24h model, a 48h model, a 72h model — each trained independently rather than one sequence model predicting all three. This avoids compounding errors that a step-by-step (recursive) forecast would build up over longer horizons — the field calls this the "direct multi-step" strategy, and it's the standard choice for exactly this reason.
4. **A tuned decision threshold:** rather than the default 50% cutoff, each horizon's threshold is chosen to catch **85% of real floods** (recall), accepting more false alarms in exchange — because in an early-warning system, a missed flood is far more costly than an extra warning.
5. **Output:** a risk level (low/moderate/high) per horizon, plus the raw probability and a plain-language "reasoning" field explaining what drove the call.



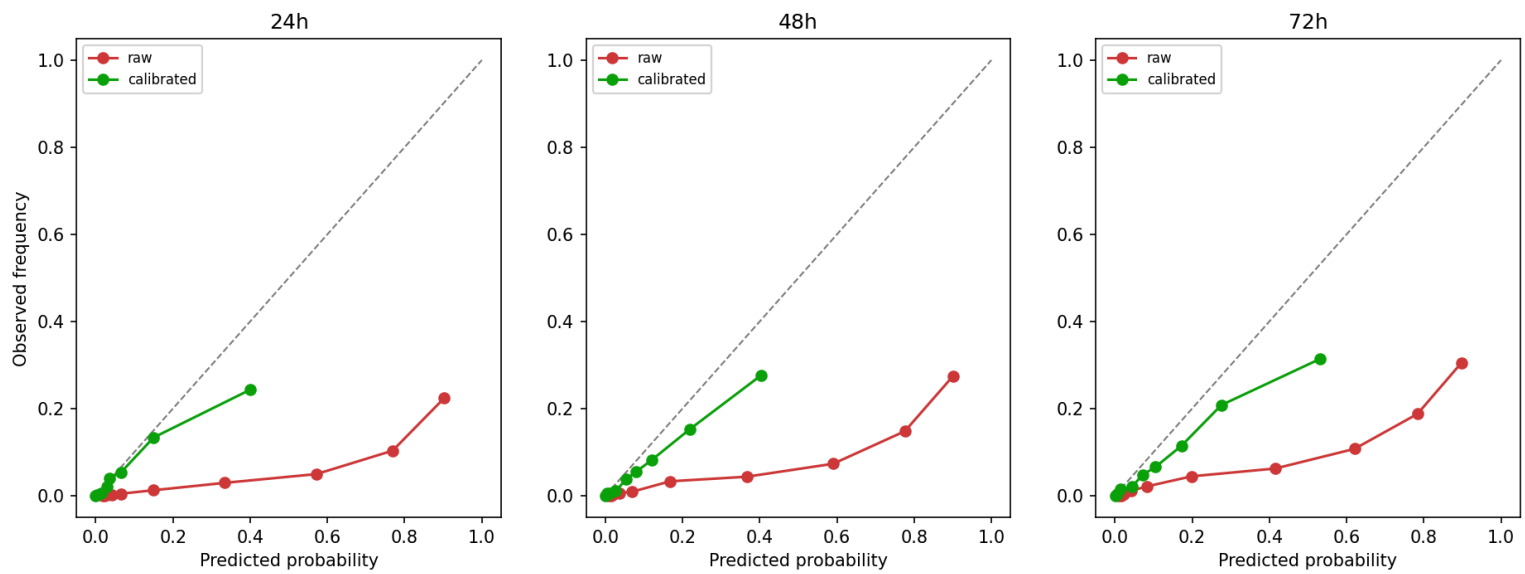
24h horizon: ROC-AUC 0.883, PR-AUC 0.214, on a genuinely held-out test set (everything after Jan 1 2024 — never seen during training).

3. The numbers you'll be asked about

Horizon	ROC-AUC	PR-AUC	Recall (tuned)	Precision (tuned)
24h	0.883	0.214	85.0%	13.6%
48h	0.864	0.245	85.0%	16.1%
72h	0.849	0.261	85.0%	18.2%

Training data: 77,960 rows (24h horizon), test set 22,890 rows — a strict **time-based split** (train = everything before Jan 1 2024, test = everything after). Never a random shuffle, because a random split would let information from a day right next to a training day leak into the test set.

Calibration (§6.3, §9 of the full report): a separate isotonic calibrator was trained per horizon and measurably fixes the raw model's overconfidence (a raw "90% chance" really only happened ~23% of the time; calibration closes most of that gap — see the figure below). **It is not yet wired into the live API** — the live `/predict` endpoint still returns the raw, uncalibrated probability. This is a disclosed, known gap, not an oversight: the decision threshold (which drives the risk level you actually see) was tuned on raw scores and works correctly regardless; only the numeric probability shown to a user would look more honest with calibration wired in.



Raw vs. calibrated reliability, all 3 horizons — three separate proper scoring rules (Brier, log loss, ECE) all agree calibration is a real improvement.

4. Likely defense questions, answered

Q1: Why LightGBM, not a neural network / LSTM? The literature is explicit that deep sequence models need large data volumes to beat gradient boosting — Google's own Flood Hub trains on thousands of gauges; this project has 30. Multiple papers also note gradient-boosted trees are specifically well-suited to rare-event, imbalanced classification like this. LightGBM matches both findings, not just convenience.

Q2: Why is precision so low (14–18%)? Deliberate, not accidental. The threshold was tuned to catch 85% of real floods, which necessarily means accepting more false alarms — a documented trade-off (§4.3), because in this context a missed flood is far worse than an extra warning. A higher-precision threshold is one config change away if a different trade-off is ever wanted.

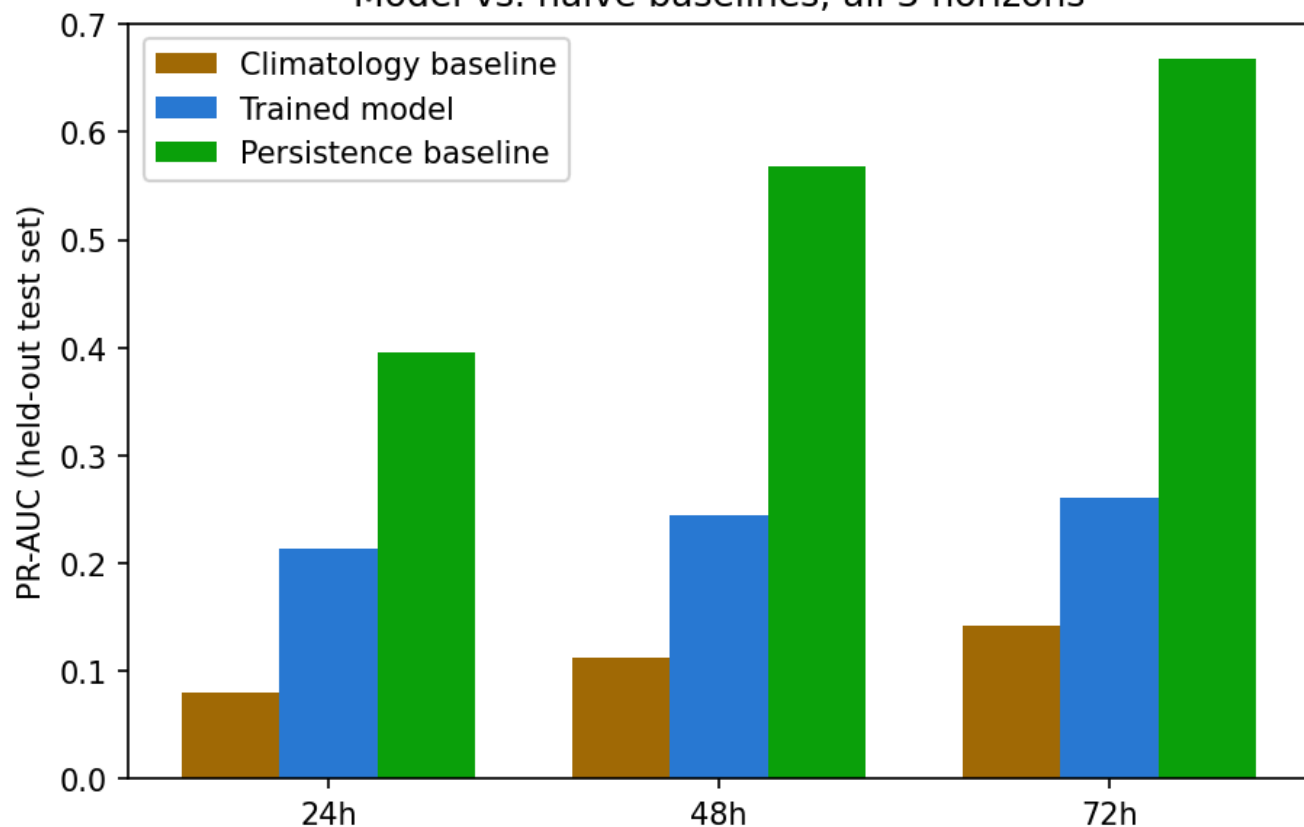
Q3: What does ROC-AUC / PR-AUC actually mean, in plain terms? - **ROC-AUC**: how well the model ranks a random flooded day above a random non-flooded day. 1.0 = perfect, 0.5 = coin flip. 0.88 means it's genuinely good at telling the two apart. - **PR-AUC**: focuses specifically on the rare positive (flood) class — a much harder bar than ROC-AUC when floods are rare. The "random guessing" baseline for PR-AUC here is roughly the true positive rate (~4% at 24h), so a PR-AUC of 0.214 is over 5x better than random, even though it "sounds" low.

Q4: How was class imbalance (floods are rare) handled? Not with a blanket `class_weight="balanced"`. A custom sample-weighting scheme was built from the real observed class balance in the most-trusted label subset, with an extra confidence discount for lower-trust label sources (event catalogs vs. direct satellite detections) — because the raw labels come from 4 sources of differing reliability, not one clean ground truth.

Q5: What does "calibration" mean and why does it matter here? A raw model's "70% probability" doesn't necessarily mean the event actually happens 70% of the time — this model's raw output was badly overconfident. Calibration is a second, separate model fit on top that corrects the *stated* probability to match *reality*, without touching the underlying ranking or the tuned decision threshold. See §3 above for its current (not-yet-live) status.

Q6: Does this model actually add value, or is it just noise dressed up as a forecast? The honest answer, stated plainly in the report's §8: on the pooled PR-AUC metric, it does **not** beat a naive "tomorrow = today" persistence baseline (0.396 vs. the model's 0.214 at 24h). The working, not-yet-proven explanation: flood labels cluster in multi-day contiguous blocks (real floods last several days), so persistence is unusually strong specifically at predicting the *continuation* of an already-ongoing flood — arguably the easiest, least-skill-requiring part of the problem. The model's real value is plausibly concentrated in predicting flood **onset** — a brand-new event persistence can never predict, since by definition it only ever repeats yesterday's label. This is flagged as the single highest-priority next investigation, not swept under the rug.

Model vs. naive baselines, all 3 horizons

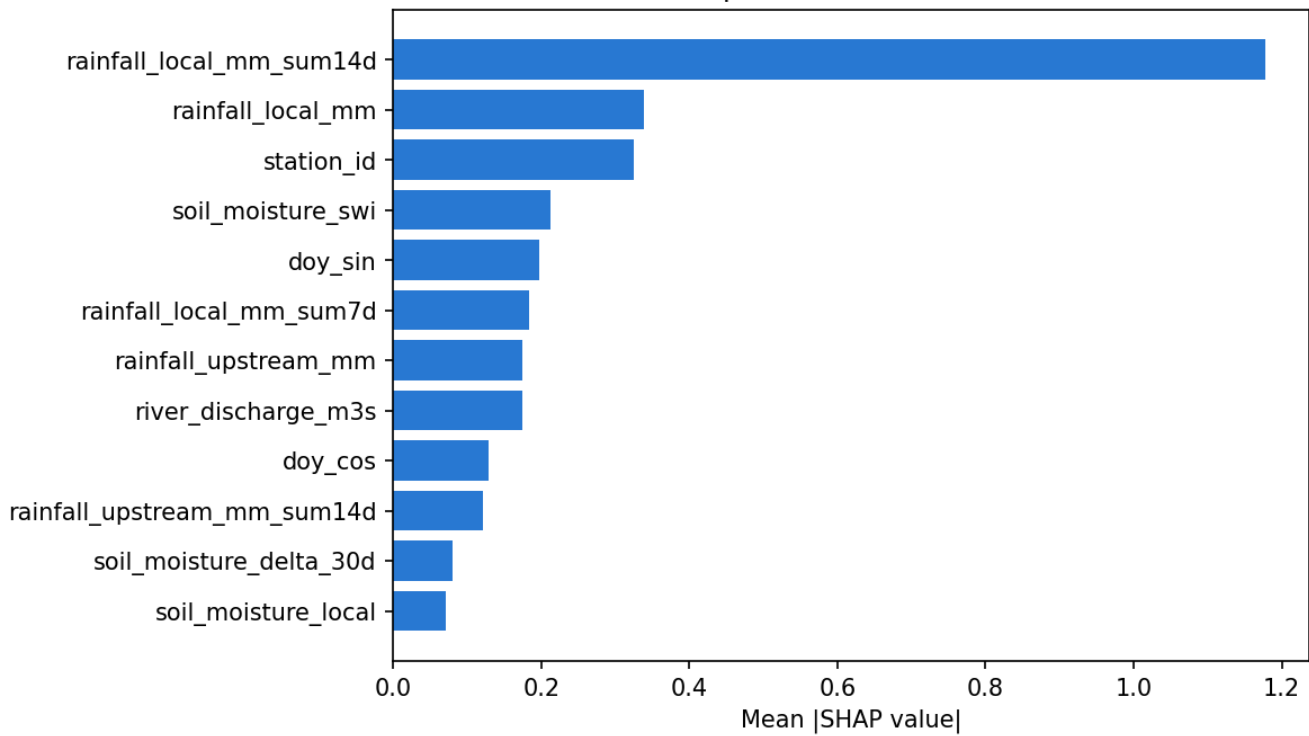


Model PR-AUC vs. climatology and persistence baselines, all 3 horizons — the honesty check from Q6.

Q7: How does this compare to the 97%+ accuracy Bangladesh studies you'll get asked about? Those numbers are almost always solving an easier problem — static classification, usually on a random (not time-based) split — not this model's genuine temporal forecasting on a strict time-ordered holdout. Accuracy is also a misleading metric here: on an imbalanced dataset, a model that always predicts "no flood" scores high on accuracy while being useless. That's exactly why this report uses PR-AUC and recall instead of accuracy — the field's own recommended metrics for this kind of problem, not a metric invented for convenience.

Q8: What's SHAP, and why is it in this report? SHAP explains *which input features drove a specific prediction* — e.g., "this alert fired mainly because of 14-day cumulative rainfall," not just a black-box number. The 2025 XAI literature frames this explicitly as a trust requirement for disaster-management adoption, not a nice-to-have visualization.

Feature importance, 24h horizon model



14-day cumulative local rainfall dominates the 24h model's decisions.

5. Weaknesses to own before they're asked

- Doesn't beat persistence on pooled PR-AUC (Q6) — the single most important limitation, already covered above so it never looks like it was hidden.
- Calibration exists but isn't live yet (§3).
- Precision is genuinely low at the tuned threshold (13–18%) — disclosed as a structural, deliberate consequence, not a bug (Q2).
- Even after calibration, the model rarely predicts very high probabilities (tops out around ~40% mean predicted probability at 24h even when calibrated) — a real, bounded improvement, not a complete fix.

6. If they only remember one thing

Three independent, time-split-evaluated LightGBM classifiers, deliberately tuned to catch 85% of real floods at the cost of more false alarms, honestly reported against naive baselines — including the one place (persistence) where it doesn't yet win, with a credible, stated reason why and a clear next step.